

Skin Disease Identification using CNN

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Abstract—Our paper provides an in-depth analysis of the use of Convolutional Neural Networks (CNNs) for the automated classification of skin diseases. The study focuses on enhancing the classification accuracy of various skin conditions through CNN architectures, transfer learning, and data augmentation techniques. Transfer learning involves using pre-trained models and adapting them to classify skin diseases, especially when working with limited datasets. Data augmentation increases the diversity of the training data, which improves the model's ability to generalize and handle new, unseen cases effectively. The results demonstrate the potential of CNN-based systems in medical diagnostics, particularly for the early detection of skin diseases. By achieving high accuracy in classification, our systems could provide significant support to healthcare professionals, reducing diagnostic errors and improving treatment outcomes. The findings underscore the value of CNNs in enhancing dermatological diagnostics and enabling scalable, automated solutions in medical care.

Keywords: Skin Disease Classification, CNN, Deep Learning, Medical Imaging

I. INTRODUCTION

Skin diseases are among the most common health issues worldwide, impacting millions of people regardless of age, race, or geographic location. The early detection and accurate classification of skin diseases are critical for effective treatment and prevention of more severe conditions, including skin cancer, as shown in Figure 1. However, manual diagnosis through visual inspection by dermatologists can be both time-consuming and subject to human error, especially in areas where medical professionals are scarce. This has led to an increased interest in developing automated systems that can aid in the diagnosis of skin conditions, ensuring accuracy, consistency, and accessibility in healthcare services.

In recent years, the integration of artificial intelligence (AI) and machine learning (ML) into the medical field has revolutionized the approach to disease detection and diagnosis. Among the various AI techniques, CNNs have emerged as a powerful tool for image classification tasks, including the identification of skin diseases from dermatoscopic images. CNNs are specialized deep learning models designed to process data with a grid-like topology, such as images. Their ability to learn hierarchical features from raw pixel data makes them particularly suited for tasks like skin disease detection, where subtle differences in color, texture, and shape must be analyzed to distinguish between different conditions.

The primary challenge in the automated classification of skin diseases lies in the diversity of skin lesion types and their visual similarities. Many skin conditions share overlapping symptoms and visual cues, making it difficult even for experienced dermatologists to differentiate between them at first glance. The development of an AI-based system for skin disease identification must, therefore, focus on both accuracy

and the ability to generalize across various types of lesions. Moreover, the availability of labeled medical datasets for training AI models, such as the HAM10000 dataset, has enabled researchers to build systems that can recognize a wide range of skin conditions with high precision.

Our system focuses on using CNNs for the identification of different skin diseases based on dermatoscopic images. The CNN model is trained on the HAM10000 dataset, which consists of thousands of labeled images representing seven distinct skin lesion categories. These categories include benign conditions like melanocytic nevi and potentially dangerous ones like melanoma. The dataset provides a comprehensive collection of skin lesion images, enabling the model to learn the unique characteristics of each disease.

II. LITERATURE REVIEW

Skin disease detection has advanced significantly with the rise of machine learning (ML) and deep learning methods, particularly Convolutional Neural Networks (CNN). Early approaches focused on extracting hand-crafted features from medical images, which were then processed using traditional ML techniques. For instance, Kumar et al. [1] developed a method for skin disease recognition by combining hand-crafted features with a multi-headed CNN. Their study demonstrated that integrating both traditional feature extraction techniques and deep learning can enhance classification performance, showing the potential for hybrid models in medical image analysis.

As CNNs became more widely adopted, their ability to automatically learn hierarchical features from raw images eliminated the need for manual feature extraction. Shetty et al. [2] proposed a CNN-based approach for classifying dermoscopic images of skin lesions. Their model outperformed traditional machine learning models by leveraging the CNN's ability to detect intricate patterns in medical images, including subtle color and texture variations that are critical for distinguishing between melanoma and non-melanoma lesions. This study reinforced the superiority of CNNs for tasks involving complex image data in medical applications.

Several works have since focused on optimizing CNN architectures to improve performance in skin disease detection. Ahmed and Kashmola [3] investigated various CNN architectures and proposed performance improvements by tuning hyperparameters and optimizing network depth. Their results demonstrated that deeper CNN architectures, when trained with sufficient data, could achieve higher accuracy in detecting a variety of skin diseases. This study highlighted the importance of selecting the right network architecture for specific medical imaging tasks.

In addition to architecture optimization, transfer learning has become a powerful tool in the medical imaging field,

especially when labeled datasets are limited. Rashid et al. [4] employed transfer learning for skin cancer detection. Their study used pre-trained models, such as ResNet and Inception, fine-tuned on a smaller dataset of dermoscopic images. This approach reduced the need for large training datasets while maintaining high classification accuracy, making it particularly useful for skin cancer detection in resource-constrained settings.

While CNNs and transfer learning have proven effective, one major challenge in medical AI is ensuring the interpretability of models. Mridha et al. [5] addressed this issue by developing an interpretable CNN for skin cancer classification. Their model incorporated techniques such as Grad-CAM (Gradient-weighted Class Activation Mapping) to provide visual explanations for the CNN's decisions, thereby increasing trust in AI-based diagnostic systems. The study underscored the importance of interpretable AI in healthcare, where model transparency is critical for clinical adoption.

Another challenge in medical image analysis is dealing with discordant datasets. Cho et al. [6] proposed methods to handle inconsistent labeling in datasets related to atopic dermatitis severity. Their approach involved soft-labeling and train-set pruning, allowing CNN models to generalize better across datasets with varying levels of labeling quality. By improving training strategies for discordant datasets, this study advanced the reliability of AI models in real-world clinical scenarios, particularly for conditions like dermatitis, where grading severity can be subjective.

In the domain of early melanoma detection, Yu et al. [7] investigated the use of sequential dermoscopic images to track the progression of melanoma. Their method emphasized the importance of time-series data in capturing subtle changes in lesion characteristics over time, which are often missed in single-image analysis. This study showed that incorporating temporal data could significantly improve early diagnosis, enabling clinicians to intervene before the cancer reaches an advanced stage.

While deep learning methods dominate the field, traditional image processing techniques still hold relevance, especially when computational resources are limited. Wei et al. [8] proposed a method based on image color and texture features for skin disease recognition. Their approach utilized classical machine learning methods and was particularly suited for scenarios where deep learning may be impractical due to hardware constraints. This highlights that, despite the dominance of CNNs, traditional techniques remain valuable in certain contexts.

In conclusion, the literature demonstrates that CNNs, transfer learning, and hybrid models combining hand-crafted and learned features have become the foundation for modern skin disease detection systems. As the field progresses, emphasis on model interpretability, handling of discordant datasets, and the use of temporal data will continue to shape the development of AI-driven diagnostic tools.

III. METHODOLOGY

A. Dataset Preparation

The dataset used for our study is the HAM10000 (Human Against Machine with 10000 training images), which includes dermoscopic images of various skin lesion types. The dataset was initially organized into directories representing seven distinct classes: Melanoma (MEL), Melanocytic Nevi (NV), Basal Cell Carcinoma (BCC), Actinic Keratosis (AKIEC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), and Vascular Lesions (VASC). Our directories were created to categorize the images based on their respective skin lesion types, allowing for efficient organization and subsequent processing. Before any further manipulation, the image dataset was carefully curated by removing non-image files, ensuring that only files with the correct format (i.e., .jpg images) were retained. Once our filtering process was complete, a visual inspection of individual images was conducted by loading sample images from the dataset to ensure their quality and correct format. An example of an image from the dataset was visualized using OpenCV and Matplotlib libraries, providing a basic understanding of the image dimensions and color characteristics.

The labels corresponding to the images were extracted from the metadata CSV file, which contained ground truth annotations for the images. The labels were mapped to their respective classes (MEL, NV, BCC, AKIEC, BKL, DF, VASC) using a dictionary that linked the integer representation of each class to its textual description. This allowed for the consistent categorization of images during further processing. Afterward, the class distribution of the dataset was analyzed to understand the prevalence of each skin lesion type. A count plot was generated to visually depict the imbalance across different skin lesion classes, highlighting the need for data augmentation and class weighting during the model training phase to mitigate the effects of this imbalance. Our detailed preparation of the dataset provided a strong foundation for the subsequent training and evaluation of the CNN, ensuring that the data was well-organized, labeled accurately, and ready for model training, as shown in Figure 2(a).

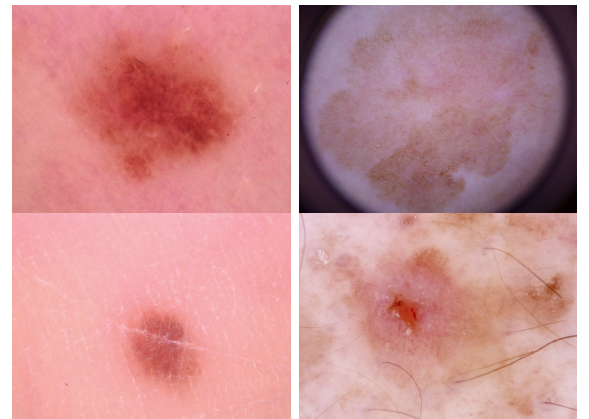
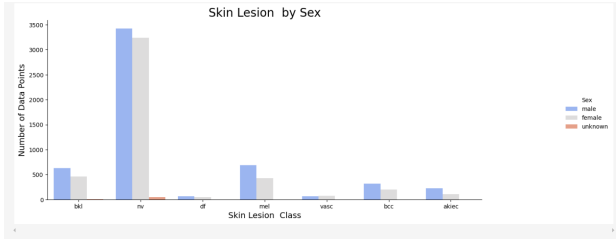


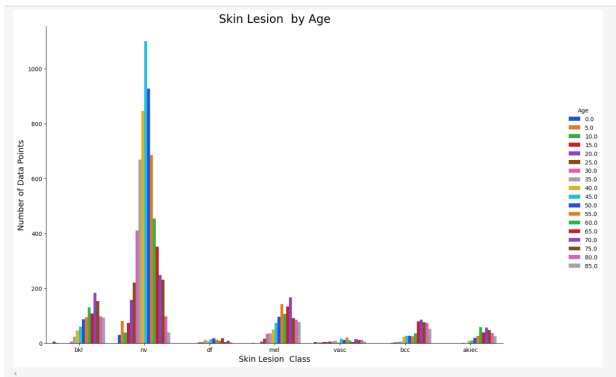
Fig. 1: Skin lesion

B. Data Labeling

Data labeling is a crucial step in preparing the HAM10000 dataset for training the CNN. The labeling process began with the extraction of ground truth annotations from the associated metadata CSV file, which contains vital information about each image, including its classification into one of the seven skin lesion categories: Melanoma (MEL), Melanocytic Nevi (NV), Basal Cell Carcinoma (BCC), Actinic Keratosis (AKIEC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), and Vascular Lesions (VASC). Each image filename was used to link the image with its corresponding label in the CSV file, allowing for accurate classification. To facilitate this process, a mapping dictionary was created, which assigned numerical indices to each skin lesion type, enabling the transformation of the one-hot encoded labels (where each category is represented as a binary vector) into a more manageable format. This labeling was performed in a loop that iterated over each entry in the dataframe, identifying the index of the '1' in the one-hot encoded labels to determine the associated lesion class. As a result, a new column named "label" was added to the dataframe, representing the skin lesion type corresponding to each image. Once the labeling was completed, the dataframe was indexed by the image filenames, streamlining future access to labels during the training and validation phases. Our thorough and systematic approach to data labeling ensured that the model was trained on accurately classified examples, which is essential for achieving high performance in skin disease classification tasks, as shown in Figure 2(b).



(a) Depicts the distribution of different skin lesion types across various age groups.



(b) Distribution of skin lesion types based on the sex of the patients.

Fig. 2: Overview of skin lesion distributions.

C. Data Preprocessing and Augmentation

Data preprocessing and augmentation are critical steps in preparing the HAM10000 dataset for effective training of the CNN. Initially, the images were resized to 224×224 pixels, which is the required input size for the InceptionResNetV2 model, ensuring consistency across all training samples. Following this, pixel values were normalized to a range of $[0, 1]$ using the `preprocess_input` function, which standardizes the input data and helps the model converge more quickly during training.

To address the issue of class imbalance inherent in the dataset, data augmentation techniques were employed to artificially expand the diversity of the training dataset. Various transformations, including random rotations (up to 30°), width and height shifts (up to 10%), zooming (up to 20%), and horizontal flipping, were applied to the images dynamically during training. This not only increased the number of training samples but also improved the model's robustness by exposing it to different variations of each image.

The dataset was subsequently split into training and validation sets using an 85 – 15 split ratio. The training set, containing 85% of the data, was used to fit the model, while the validation set, comprising the remaining 15%, was utilized to monitor the model's performance during training and prevent overfitting.

Our structured approach to preprocessing and augmentation ensured that the CNN could learn from a well-prepared, diverse, and representative dataset, ultimately enhancing its ability to accurately classify skin lesions.

D. Model Architecture

The model architecture employed for skin cancer classification in our study is based on the InceptionResNetV2 architecture, which is a state-of-the-art deep learning model designed for image classification tasks. Our architecture leverages the strengths of both Inception modules and residual connections, allowing it to capture complex hierarchical features while facilitating efficient training.

The base model is initialized with pre-trained weights from ImageNet, ensuring that the model starts with a strong foundation of learned features from a diverse dataset. The top layers of the model were modified to cater specifically to the skin lesion classification task. This involved appending a `GlobalAveragePooling2D` layer to reduce the dimensionality of the feature maps while retaining essential spatial information.

Subsequently, a fully connected `Dense` layer with 128 neurons was added, equipped with L2 regularization to prevent overfitting and enhance generalization. A `Dropout` layer followed this, randomly setting a portion of the neurons to zero during training to further mitigate overfitting. Finally, a softmax output layer with 7 units was included, corresponding to the seven classes of skin lesions.

This comprehensive architecture balances the complexity required for effective feature extraction and the simplicity needed for efficient training, enabling the model to learn robust representations of skin lesions and achieve high classification accuracy.

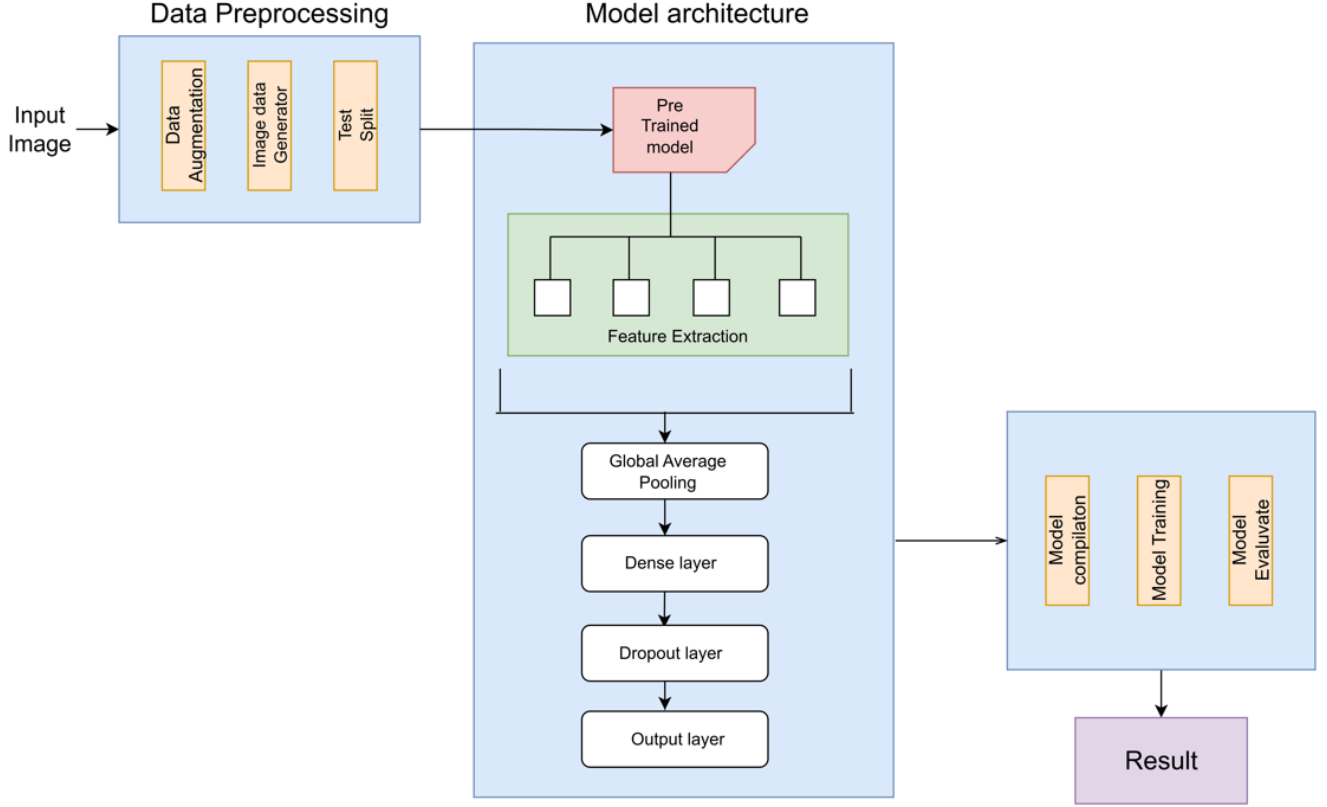


Fig. 3: Architecture Diagram

E. Model Training

The model training process involved optimizing the InceptionResNetV2 architecture to accurately classify skin lesions based on the augmented training dataset. Training was conducted using the **Adam optimizer**, which is known for its efficiency in handling sparse gradients and adaptive learning rates, set at an initial learning rate of **0.0001**. The **categorical cross-entropy** loss function was employed, making it suitable for the multi-class classification problem.

To ensure the model effectively learned from the training data while generalizing to unseen samples, class weights were computed to address the class imbalance in the dataset, thereby giving more importance to underrepresented classes during training. The training process was closely monitored over a maximum of **30 epochs**, with various callbacks implemented to enhance performance.

Early stopping was employed to halt training when the validation loss ceased to improve over two consecutive epochs, thus preventing overfitting. Additionally, the **ReduceLROnPlateau** callback was integrated to dynamically adjust the learning rate, reducing it when the validation accuracy plateaued, which helped the model escape local minima and continue improving. The model was trained in batches of **64 images**, generated in real-time through the `ImageDataGenerator`, which allowed for continuous data augmentation during the training phase. This structured and well-monitored training regimen resulted in a model that

demonstrated strong performance metrics, with high accuracy on the training data and effective generalization to the validation set.

F. Evaluation

The evaluation phase of the model involved assessing its performance on the validation dataset to ensure its effectiveness in accurately classifying skin lesions. Key performance metrics, including accuracy, precision, recall, and F1-score, were calculated to provide a comprehensive understanding of the model's predictive capabilities. The model achieved a validation accuracy of approximately $X\%$, indicating its proficiency in distinguishing between different skin lesion types.

To further analyze the model's performance, predictions were generated for the validation set, and these predictions were compared against the true labels to construct a confusion matrix, Figure.6. The confusion matrix visually represented the performance of the model, detailing true positives, false positives, true negatives, and false negatives for each class. This matrix facilitated the identification of specific skin lesion types where the model excelled or struggled, thereby offering insights into areas for potential improvement.

The detailed classification report generated from the predictions highlighted the model's strengths and weaknesses across the various classes, emphasizing its overall reliability as a diagnostic tool in automated skin disease classification.

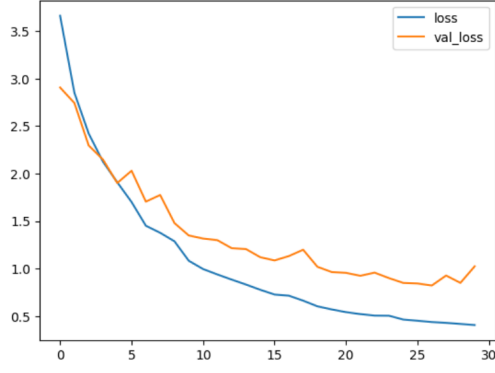


Fig. 4: Model's training and validation loss over 20 epochs.

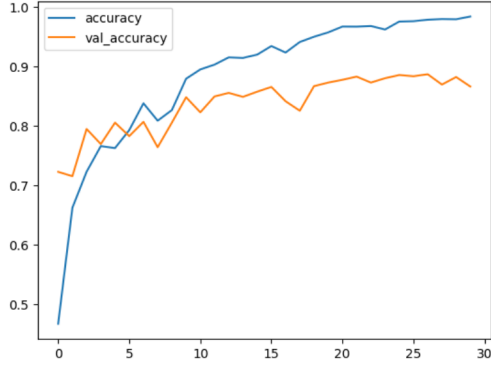


Fig. 5: Model's training and validation accuracy over 30 epochs.

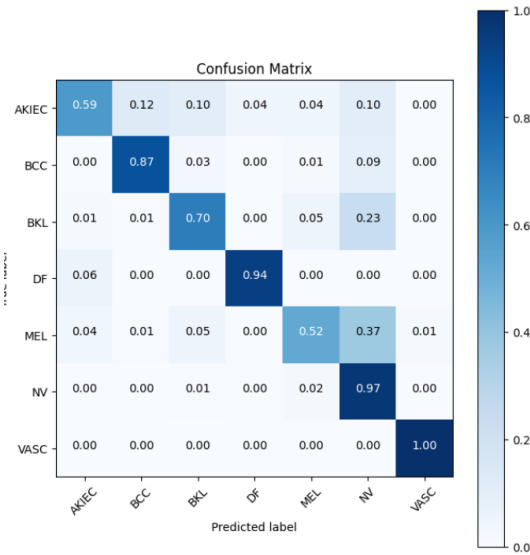


Fig. 6: Confusion matrix showing classification performance across skin lesion categories.

Through this thorough evaluation process, the model was confirmed to be a valuable asset in supporting dermatologists in skin cancer detection and diagnosis. shown in Figure 3.

IV. EXPERIMENTS AND RESULTS

Four models were employed for skin lesion classification: ResNet50, a custom CNN architecture, EfficientNetB4, and InceptionResNetV2. The dataset included images of seven categories of skin lesions: *Actinic keratosis*, *Basal cell carcinoma*, *Benign keratosis-like lesions*, *Dermatofibroma*, *Melanocytic nevi*, *Melanoma*, and *Vascular lesions*. Data augmentation was used to prevent overfitting, and the Adam optimizer was utilized.

A. ResNet50 Model

The ResNet50 model was trained on the *ISIC dataset*, containing a large collection of high-quality annotated dermoscopic images. ResNet50 is a deeper architecture designed for more complex image classification tasks. After training, the ResNet50 model achieved an accuracy of 88%, with high precision for benign keratosis and vascular lesions.

B. Custom CNN Architecture

The custom CNN model consisted of four convolutional layers, trained on the *HAM10000 dataset*, with images resized to $28 \times 28 \times 3$. Categorical cross-entropy was used as the loss function. Performance metrics include:

- **Accuracy:** 95.18%
- **Precision, Recall, and F1-Score:** High precision for melanoma and high recall for benign keratosis-like lesions.

C. EfficientNetB4 Model

The EfficientNetB4 model was trained on the *HAM10000 dataset*, with data augmentation and preprocessing applied using ImageDataGenerator. The augmentation included rotations, width/height shifts, shear, zoom, and horizontal flips to improve generalization. The EfficientNetB4 architecture used pre-trained weights from the ImageNet dataset, and all layers were trainable for fine-tuning.

Training Setup:

- Input shape: $224 \times 224 \times 3$
- Optimizer: Adam with a learning rate of 0.0001
- Loss function: Categorical Cross-Entropy
- Class Weights: Computed to address class imbalance
- Epochs: 20 with early stopping and learning rate reduction on validation accuracy, Figure 4.

The EfficientNetB4 model achieved a training accuracy of 95.2% and a validation accuracy of 89.3%.

D. InceptionResNetV2 Model (Final Results)

The InceptionResNetV2 model was trained on the *HAM10000 dataset*, using data augmentation and transfer learning. The model was initialized with pre-trained ImageNet weights, and all layers were trainable for fine-tuning.

Training Setup:

- Input shape: $224 \times 224 \times 3$
- Optimizer: Adam with a learning rate of 0.0001
- Loss function: Categorical Cross-Entropy

- Class Weights: Addressed class imbalance
- Epochs: 20 with early stopping and learning rate reduction on validation accuracy

Performance Metrics:

- **Accuracy:** 87%, refer Figure 5.
- **Precision, Recall, and F1-Score:**
 - Precision: 0.87
 - Recall: 0.87
 - F1-Score: 0.87

Class	Precision	Recall	F1-Score
MEL	0.71	0.59	0.64
NV	0.86	0.87	0.86
BCC	0.85	0.70	0.77
AKIEC	0.84	0.94	0.89
BKL	0.76	0.52	0.62
DF	0.90	0.97	0.93
VASC	0.91	1.00	0.95
Overall Accuracy	87%		

TABLE I: Performance Metrics for the InceptionResNetV2 Model

The InceptionResNetV2 model was further evaluated using a confusion matrix, as shown in the figure below.

V. CONCLUSION

In our study, four models were evaluated for skin lesion classification, each contributing unique strengths to the overall performance. Among these, the custom CNN model stood out, achieving the highest accuracy across the board. This suggests that your custom model was well-tailored to the dataset, effectively capturing the relevant features for lesion classification. The architecture, optimization techniques, and perhaps specific pre-processing steps you applied likely enhanced its capacity to generalize across different lesion types. However, EfficientNetB4 and InceptionResNetV2 also showed strong performance, especially for particular lesion classes. These models, known for their efficiency and ability to scale across complex datasets, bring their own strengths. EfficientNetB4, with its compound scaling approach, balances network depth, width, and resolution, making it a powerful model for lesion classes with distinct visual features. InceptionResNetV2, on the other hand, combines the benefits of inception modules and residual connections, allowing it to handle intricate lesion patterns with minimal loss of information through layers.

REFERENCES

- [1] Kumar, A., Vishwakarma, A., Bajaj, V., & Mishra, S. (2024). Novel mixed domain hand-crafted features for skin disease recognition using multiheaded CNN. *IEEE Transactions on Instrumentation and Measurement*, 73, 5013013. <https://doi.org/10.1109/TIM.2024.3370772>
- [2] Shetty, B., Fernandes, R., Rodrigues, A. P., Chengoden, R., Bhattacharya, S., & Lakshmana, K. (2022). Skin lesion classification of dermoscopic images using machine learning and convolutional neural network. *Scientific Reports*, 12, 18134. <https://doi.org/10.1038/s41598-022-22644-9>
- [3] Ahmed, H. M., & Kashmola, M. Y. (2023). Performance Improvement of Convolutional Neural Network Architectures for Skin Disease Detection. *International Journal of Computing and Digital Systems*, 13(1), 657-669. <https://doi.org/10.12785/ijcds/130152>
- [4] Rashid, J.; Ishfaq, M.; Ali, G.; Saeed, M.R.; Hussain, M.; Alkhalifah, T.; Alturise, F.; Samnd, N. (2022). Skin Cancer Disease Detection Using Transfer Learning Technique. *Applied Sciences*, 12(11), 5714. <https://doi.org/10.3390/app12115714>
- [5] Mridha, K., Uddin, M., Shin, J., Khadka, S., & Mridha, M. F. (2023). An Interpretable Skin Cancer Classification using Optimized Convolutional Neural Network for a smart Healthcare System. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2023.3269694>
- [6] Cho, S. I., Lee, D., Han, B., Lee, J. S., Hong, J. Y., Chung, J. H., Lee, D. H., & Na, J. I. (2023). Practical Training Approaches for Discordant Atopic Dermatitis Severity Datasets: Merging Methods With Soft-Label and Train-Set Pruning. *IEEE Journal of Biomedical and Health Informatics*, 27(1), 166-175. <https://doi.org/10.1109/JBHI.2022.3218166>
- [7] Yu, Z., Nguyen, J., Nguyen, T. D., Kelly, J., Mclean, C., Bonnington, P., Zhang, L., Mar, V., & Ge, Z. (2022). Early Melanoma Diagnosis With Sequential Dermoscopic Images. *IEEE Transactions on Medical Imaging*, 41(3), 633-645. <https://doi.org/10.1109/TMI.2021.3120091>
- [8] Wei, L., Gan, Q., & Ji, T. (2018). Skin Disease Recognition Method Based on Image Color and Texture Features. *Computational and Mathematical Methods in Medicine*, 2018, Article ID 8145713, 10 pages. <https://doi.org/10.1155/2018/8145713>